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ABSTRACT

Crop analytics usually joins separate tools by hand, one analysis at a time. The Crop Analytics Orchestra is a set of 20 R packages and an open data catalogue, DataHub, built to work together. Each package does one job well on its own. All of them return results in one shared format, and each can decline to answer when the data cannot support an answer. The decision packages will not price an answer that was declined earlier. On a synthetic barley belt built on real climate and soil records, the full chain lost 49% less money on nitrogen decisions than an approach that trusts the crop simulator alone.

1 How the packages work together

- One result format.** Every package writes its result in the same structure, defined by the orchestraManifest package. The next package can check the result and use it directly.
- Declining to answer.** A package that cannot answer soundly says so, with its reason, in the same format. Later packages carry that forward instead of guessing.
- conductoR runs the plan.** An analysis plan is a list of steps, each done by one package. conductoR runs the steps in order. Before a result moves to the next step, conductoR checks that it is the right kind of result, and stops the plan if it is not.
- One install.** Seven packages are public today and install in one call:

```
install.packages(c("masque", "PESTO", "kernR",  
  "proxymix", "gpfield", "decideR", "grainPlan"),  
  repos = c("https://max578.r-universe.dev",  
    "https://cloud.r-project.org"))
```

MEET THE PACKAGES

Data and privacy. **terroir** fetches climate, weather, soil, elevation and farm-economics data for any location, and records where every value came from. **masque** makes a synthetic copy of a confidential table that keeps its experimental design, so an analysis can be built before anyone sees the real data. **DataHub** holds 81 open datasets, 15 of them Australian, each with its source and licence, and finds the datasets that can answer a given question. It is a catalogue, not an R package.

Crop simulation. **apsimR** runs the APSIM Next Generation crop simulator from R. **PESTO** fits a simulator to field observations with the ensemble methods of PEST++, and keeps what remains uncertain.

Causal, spatial and time series. **kernR** tests whether a treatment changes the whole distribution of an outcome, not only its mean. **cdzoo** puts many causal-discovery algorithms behind one interface and suggests one for the data. **bacipair** compares treated sites with matched control sites, before and after a change. **gpfield** maps a field variable from point samples to paddock or regional averages. **kalmix** does Kalman filtering, regime switching and change-point detection on series over time, and checks its causal information rate against the method authors' acir package.

Statistics and consensus. **flexyBayes** fits Bayesian mixed models for multi-environment trials, written in ASReml or brms syntax. **koine** adds a fourth opinion built from a battery of inference methods, and stays silent when it adds nothing. **quorum** combines independent estimates of one quantity into a verdict, or reports that they disagree. **survkit** analyses time-to-event data through one interface.

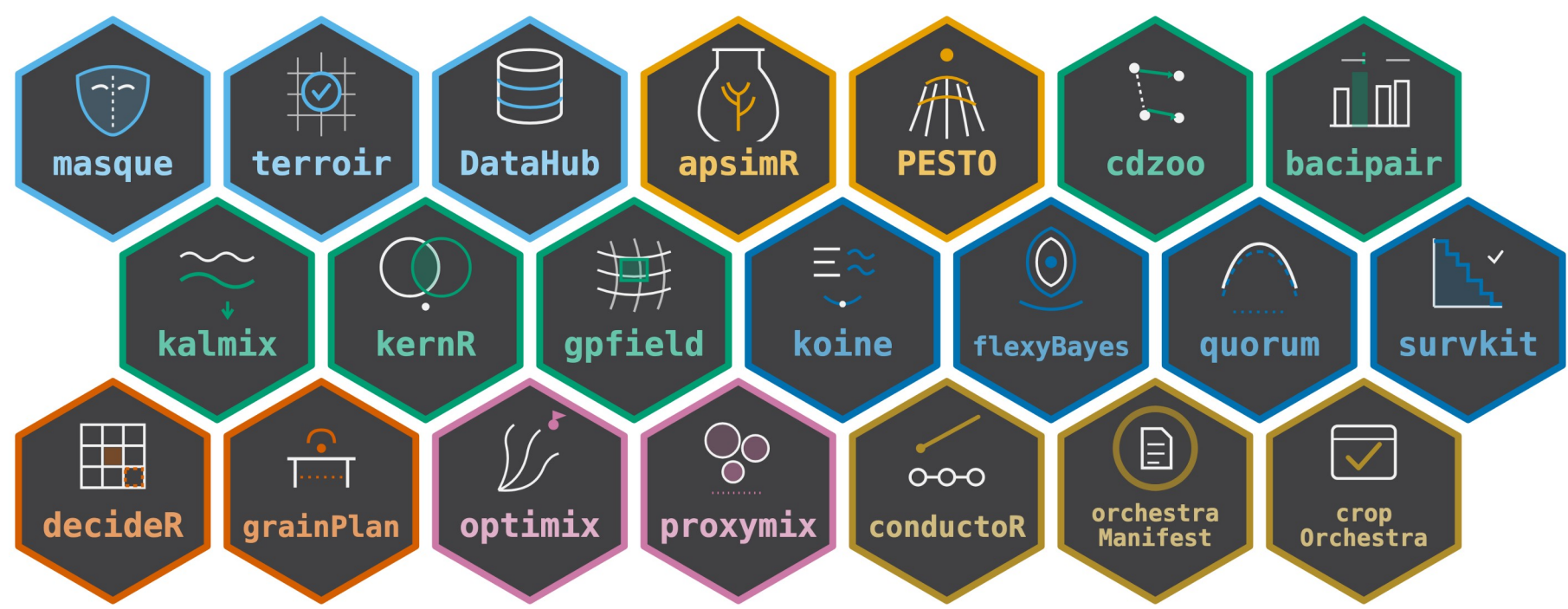
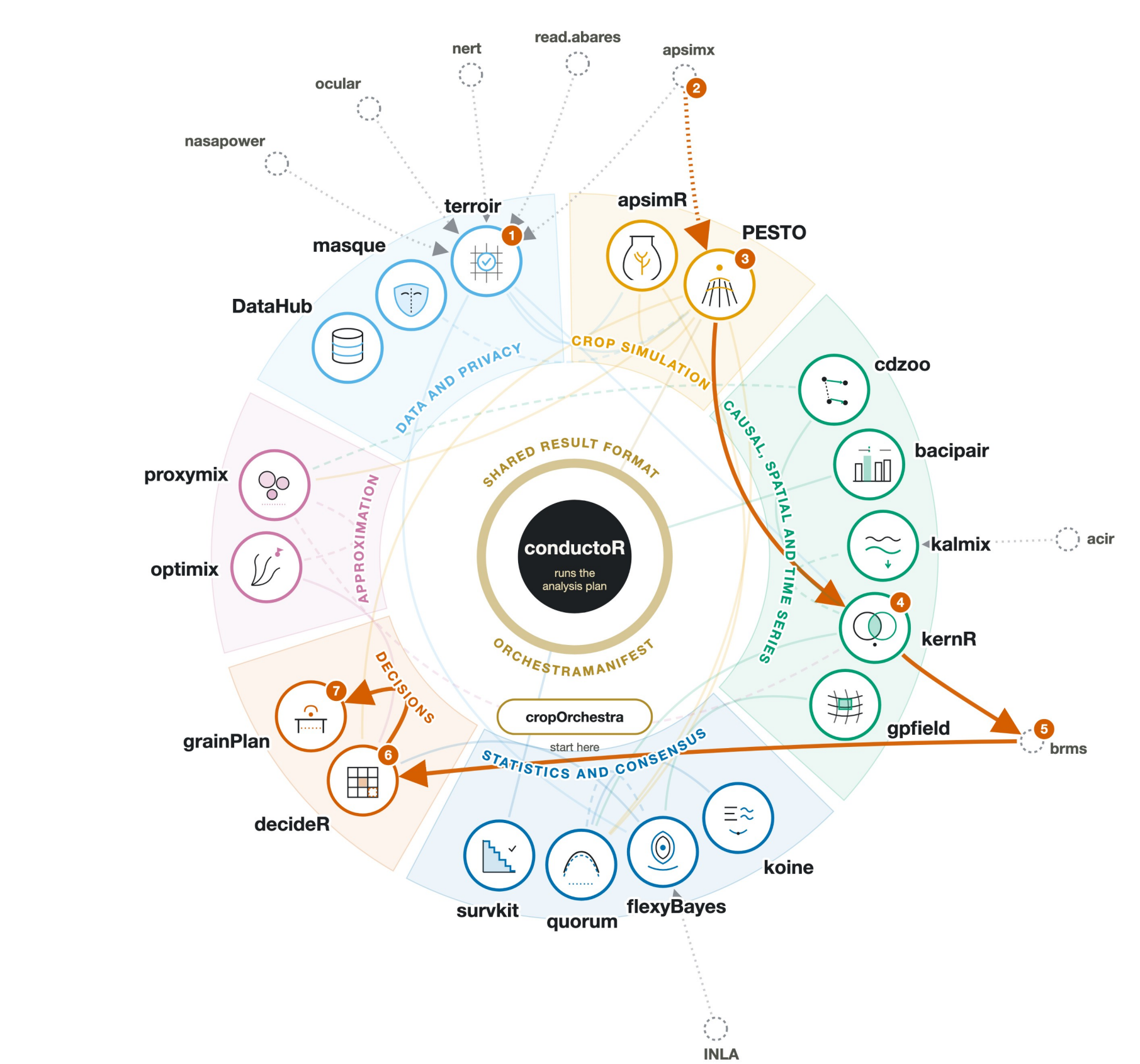
Approximation and optimisation. **proxymix** replaces a complicated distribution with a small mixture of bell curves and states the information lost. **optimix** gives one problem to many optimisers and picks the best for it.

Decisions. **decideR** chooses the best action under uncertainty and prices what more information is worth. **grainPlan** turns an analysis into a grower's plan: nitrogen rate, variety and grade target.

At the centre. **conductoR** runs an analysis plan step by step. **orchestraManifest** is the result format every package writes. **cropOrchestra** installs the set and checks it works together.

Sources: each package's DESCRIPTION, read 2026-10-01; DataHub's catalogue.

2 The orchestra



Each hex is one member: 20 R packages and DataHub, the data catalogue. Its border is the colour of its section on the ring. At the centre, conductoR runs a plan across the packages, orchestraManifest is the result format they all write, and cropOrchestra installs the lot. Dotted outer nodes are packages other people maintain. The orange numbers mark the steps of section 3.

Scan the QR code to open this map. Select any package to read what it does and how it connects.

3 One nitrogen decision, told step by step

The data: what is real and what is simulated. Real: 40 site locations across the Australian grain belt, the daily climate at each from NASA POWER for the 5 seasons 2016 to 2020, and each site's soil profile from SoilGrids. Simulated: every plot record. APSIM grows barley at each site and season at 6 nitrogen rates from 0 to 150 kg N/ha, with 3 varieties (Compass, Spartacus CL and Hindmarsh) and 3 replicates in a field of 9 rows by 6 columns: 10,800 plots in all. Each plot record holds its design (site, season, variety, rate, replicate, row and column), grain yield (kg/ha) and grain protein (%), with a field trend and measurement noise added to the simulator's output.

Because the record is simulated, the truth is known exactly: at 12 sites a hidden subsoil constraint caps yield at 57 to 77% of potential and flattens the response above 60 kg N/ha, and the simulator used in the analysis is never told.

The steps, numbered as on the map:

- terroir collects the climate record for every site and keeps its source. Soil profiles come from SoilGrids through apsimx.
- apsimx writes each site's weather file from NASA POWER and runs APSIM barley for every site, season, variety and rate. A smaller set of runs trains a fast stand-in for the simulator, an emulator built with quorum, so that it can be calibrated quickly.
- PESTO calibrates the emulator to the field record at the low rates only, 0 to 60 kg N/ha, and keeps the calibrated parameters as an ensemble of plausible values.
- kernR tests, at the higher rates, whether the field record still agrees with the calibrated model. It rejects the model at 7 of the 12 constrained sites, and wrongly at 2 unconstrained sites.
- brms fits the field record directly with plateau curves, linear and quadratic, weighted by how well each predicts plots held out of the fit. Wherever kernR rejected the model, this field-record fit is used instead.
- decideR prices every candidate rate by expected profit: grain price times yield, less nitrogen cost and a nitrous-oxide shadow price, averaged over the uncertainty. It also prices what one soil test or one strip trial at each site would be worth.
- grainPlan writes one nitrogen rate per site, with the reasons, and keeps every rejection made along the way.

The comparison arm trusts the calibrated simulator everywhere. Both arms are scored by profit foregone against the truth-optimal rate, in AUD/ha, and the whole chain is run again on a re-seeded belt.

Cause and effect. A nitrogen decision needs the yield gained by applying a rate. Here every site gets every rate, so a plain comparison measures that gain. In farm records it does not: wetter sites tend to get more nitrogen and also yield more. To show the error, 50 farm-style samples were drawn from the same plots, with more nitrogen at wetter sites, and the gain was read three ways.

4 Results

- 2,960 vs 5,818 AUD/ha** total loss against the planted truth, orchestra chain against the simulator-only approach, 49% less. The same result held on the re-seeded belt (2,549 vs 5,002).
- 1,440 kg N not applied**, 6.2 t CO₂-e/ha avoided, at the constrained sites where extra nitrogen does not pay.
- 164 AUD/ha** the value of one more strip trial at the best site, which shows where the next research dollar pays.
- 2.79 vs 1.32 t/ha** yield gained from 150 kg N/ha in the farm-style samples, read naively, against the planted gain. Adjusted for rainfall, 1.33 t/ha; with proxymix's mixture model, 1.34 (medians of 50 samples). At other rates errors reach 0.30 t/ha, so the check set beforehand, within 15% at every rate, failed.

The belt is synthetic: the numbers show the method at work, not what a real paddock would earn.

5 Get it and explore

Seven packages are open source (MIT); the others are in development. Map and every package: max578.github.io/orchestra/members (QR code). The study: max578.github.io/orchestra/a0.

6 References

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Holzworth, D., Huth, N. I., Fainges, J., Brown, H., Zurcher, E., Cichota, R., Verrall, S., Herrmann, N. I., Zheng, B., & Snow, V. (2018). APSIM Next Generation: Overcoming challenges in modernising a farming systems model. *Environmental Modelling & Software*, 103, 43–51. <https://doi.org/10.1016/j.envsoft.2018.02.002>

Sparks, A. (2018). nasapower: A NASA POWER global meteorology, surface solar energy and climatology data client for R. *Journal of Open Source Software*, 3(30), 1035. <https://doi.org/10.21105/joss.01035>

